



Maestría en Inteligencia Artificial

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Introducción a la Inteligencia Artificial

02 Agentes

mi_cueva_4x4.yaml

```
→ python3 01_wumpus_world.py --config config/mi_cueva_4x4.yaml
Step 0   Score 0   IN CAVE
4 | P   .   .   .
3 | .   .   P   .
2 | .   .   P   .
1 | >   .   W   G
      1   2   3   4
Percept [None]

Legend: >^v< agent  P pit  W wumpus  w dead wumpus  G gold  . empty  ## wall
Loaded config/mi_cueva_4x4.yaml (4x4)
```

mi_cueva_dificil_4x4.yaml

```
→ python3 01_wumpus_world.py --config config/mi_cueva_dificil_4x4.yaml
Step 0   Score 0   IN CAVE
4 | G   P   .   .
3 | .   .   P   .
2 | P   W   .   .
1 | >   .   .   .
      1   2   3   4
Percept [Breeze]

Legend: >^v< agent  P pit  W wumpus  w dead wumpus  G gold  . empty  ## wall
Loaded config/mi_cueva_dificil_4x4.yaml (4x4)
```

mi_cueva_4x4.yaml

```
Action: TurnLeft
Step 197 Score -197.0 IN CAVE
4 | P . . .
3 | v . P .
2 | . . P .
1 | . . W G
   1 2 3 4
Percept [Breeze]
Reward -1 (ok)

Action: TurnLeft
Step 198 Score -198.0 IN CAVE
4 | P . . .
3 | > . P .
2 | . . P .
1 | . . W G
   1 2 3 4
Percept [Breeze]
Reward -1 (ok)

Action: TurnLeft
Step 199 Score -199.0 IN CAVE
4 | P . . .
3 | ^ . P .
2 | . . P .
1 | . . W G
   1 2 3 4
Percept [Breeze]
Reward -1 (ok)

Action: TurnLeft
Step 200 Score -200.0 IN CAVE
4 | P . . .
3 | < . P .
2 | . . P .
1 | . . W G
   1 2 3 4
Percept [Breeze]
Reward -1 (max_steps)

Result: stopped without gold steps=200 score=-200.0
~/Desktop/wumpus via 🐍 3.14.7 via wumpus
➔
```

04_goal_based_agent.py

```
Action: Forward
Step 22 Score -32.0 IN CAVE
4 | P . . .
3 | . . P .
2 | . . P .
1 | . . < .
   1 2 3 4
Percept [Breeze]
Reward -1 (moved)

Action: Forward
Step 23 Score -33.0 IN CAVE
4 | P . . .
3 | . . P .
2 | . . P .
1 | . . < w .
   1 2 3 4
Percept [None]
Reward -1 (moved)

Action: Forward
Step 24 Score -34.0 IN CAVE
4 | P . . .
3 | . . P .
2 | . . P .
1 | < . w .
   1 2 3 4
Percept [None]
Reward -1 (moved)

Action: Climb
Step 25 Score 965.0 CLIMBED
4 | P . . .
3 | . . P .
2 | . . P .
1 | . . w .
   1 2 3 4
Percept [None]
Reward +999 (climbed_with_gold)

Result: climbed with gold steps=25 score=965.0
~/Desktop/wumpus via 🐍 3.14.7 via wumpus
➔
```

05_utility_based_agent.py

```
+ python3 06_learning_agent.py --config config/mi_cueva_4x4.yaml
Trained 1500 episodes on config/mi_cueva_4x4.yaml
Mean score (all): -26.2
Mean score (last 50): -1.7
Q-table size: 564 state-action pairs

— Greedy demo (epsilon = 0) —

Step 0 Score 0 IN CAVE
4 | P . . .
3 | . . P .
2 | . . P .
1 | > . W G
   1 2 3 4
Percept [None]
Agent: learning

Action: Climb
Step 1 Score -1.0 CLIMBED
4 | P . . .
3 | . . P .
2 | . . P .
1 | . . W G
   1 2 3 4
Percept [None]
Reward -1 (climbed_empty)

Result: climbed without gold steps=1 score=-1.0
~/Desktop/wumpus via 🐍 3.14.7 via wumpus
➔
```

06_learning_agent.py

mi_cueva_dificil_4x4.yaml

```
Action: TurnLeft
Step 196 Score -196.0 IN CAVE
 4 | G P . .
 3 | . . P .
 2 | P W . .
 1 | > . . .
   1 2 3 4
Percept [Breeze]
Reward -1 (ok)

Action: TurnLeft
Step 197 Score -197.0 IN CAVE
 4 | G P . .
 3 | . . P .
 2 | P W . .
 1 | ^ . . .
   1 2 3 4
Percept [Breeze]
Reward -1 (ok)

Action: TurnLeft
Step 198 Score -198.0 IN CAVE
 4 | G P . .
 3 | . . P .
 2 | P W . .
 1 | < . . .
   1 2 3 4
Percept [Breeze]
Reward -1 (ok)

Action: TurnLeft
Step 199 Score -199.0 IN CAVE
 4 | G P . .
 3 | . . P .
 2 | P W . .
 1 | v . . .
   1 2 3 4
Percept [Breeze]
Reward -1 (ok)

Action: TurnLeft
Step 200 Score -200.0 IN CAVE
 4 | G P . .
 3 | . . P .
 2 | P W . .
 1 | > . . .
   1 2 3 4
Percept [Breeze]
Reward -1 (max_steps)

Result: stopped without gold steps=200 score=-200.0
~/Desktop/wumpus via 🐍 3.14.7 via wumpus
+ |
```

04_goal_based_agent.py

```
Action: Forward
Step 15 Score -25.0 IN CAVE
 4 | G P . .
 3 | . . P ^
 2 | P W . .
 1 | . . . .
   1 2 3 4
Percept [Breeze]
Reward -1 (moved)

Action: Forward
Step 16 Score -26.0 IN CAVE
 4 | G P . ^
 3 | . . P .
 2 | P W . .
 1 | . . . .
   1 2 3 4
Percept [None]
Reward -1 (moved)

Action: TurnLeft
Step 17 Score -27.0 IN CAVE
 4 | G P . <
 3 | . . P .
 2 | P W . .
 1 | . . . .
   1 2 3 4
Percept [None]
Reward -1 (ok)

Action: Forward
Step 18 Score -28.0 IN CAVE
 4 | G P < .
 3 | . . P .
 2 | P W . .
 1 | . . . .
   1 2 3 4
Percept [Breeze]
Reward -1 (moved)

Action: Forward
Step 19 Score -1029.0 DEAD
 4 | G < . .
 3 | . . P .
 2 | P W . .
 1 | . . . .
   1 2 3 4
Percept [None]
Reward -1001 (fell_in_pit)

Result: died without gold steps=19 score=-1029.0
~/Desktop/wumpus via 🐍 3.14.7 via wumpus
+ |
```

05_utility_based_agent.py

```
+ python3 06_learning_agent.py --config config/mi_cueva_muy_dificil_4x4.yaml
Trained 1500 episodes on config/mi_cueva_muy_dificil_4x4.yaml
Mean score (all): -24.2
Mean score (last 50): -21.7
Q-table size: 270 state-action pairs

--- Greedy demo (epsilon = 0) ---

Step 0 Score 0 IN CAVE
 4 | G P . .
 3 | . . P .
 2 | P W . .
 1 | > . . .
   1 2 3 4
Percept [Breeze]
Agent: learning

Action: Climb
Step 1 Score -1.0 CLIMBED
 4 | G P . .
 3 | . . P .
 2 | P W . .
 1 | . . . .
   1 2 3 4
Percept [Breeze]
Reward -1 (climbed_empty)

Result: climbed without gold steps=1 score=-1.0
~/Desktop/wumpus via 🐍 3.14.7 via wumpus
+ |
```

06_learning_agent.py